

PORTFOLIO CASE STUDY

AI Founder Intelligence & Lead Enrichment Platform

Automating founder discovery and company intelligence through workflow automation, search APIs, and intelligent data enrichment.

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Workflow Automation • n8n • Serper APIs • Business Intelligence

Turning a manual research task into a scalable intelligence engine

Founder and leadership information is publicly available but every recruiting, sales, and research team re-collects it manually, company by company. This platform automates the entire discovery-to-database workflow end to end.



The Problem

Finding founder and co-founder LinkedIn profiles for 1,000 companies requires researching approximately 2,000 individuals. At an average of 2–3 minutes per profile, this represents 60–100 hours of repetitive manual work that can be standardized and automated.



The Solution

An n8n-powered automation that reads company names, searches Google through the Serper API, extracts founder and co-founder LinkedIn profiles, and automatically updates a centralized Google Sheets database.



The Impact

Based on the measured runtime of 1 minute 2 seconds for 100 companies, the workflow is expected to process approximately 1,000 companies in around 10 minutes under similar conditions.

THE BUSINESS PROBLEM

The data isn't missing. Collecting it manually is what doesn't scale.

Founder and leadership data is publicly available online — but every team that needs it repeats the same manual research from scratch, for every single company.

Open Google. Search the company. Search again for the founder. Open results. Verify the person. Find LinkedIn. Copy the URL. Paste into a spreadsheet. Then repeat — hundreds of times.

Six different roles run this exact same process today.



Recruiters



Sales Teams



Business Development



Investors



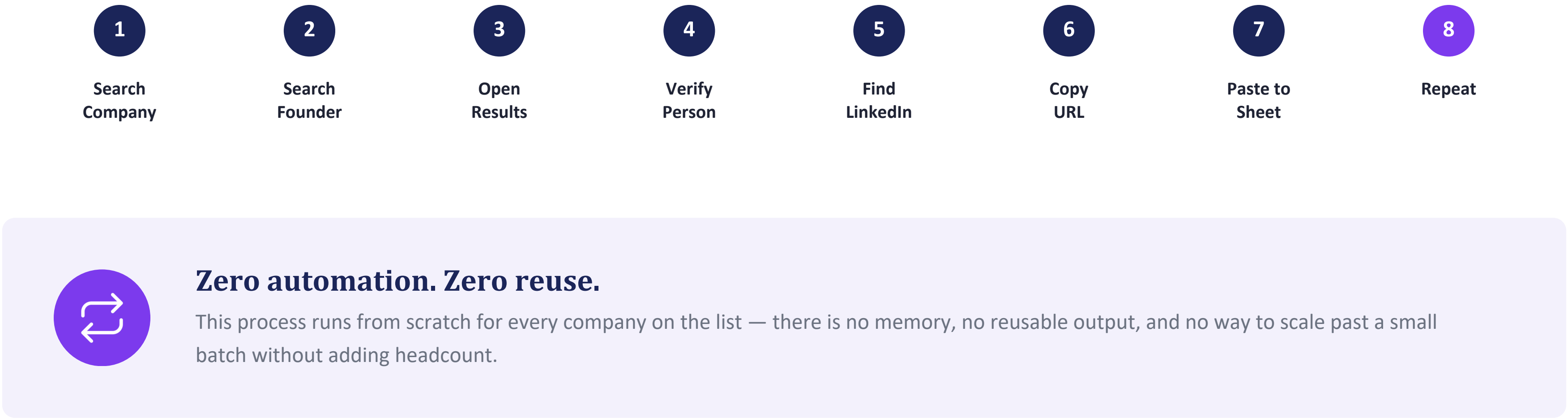
Job Seekers



Market Research

CURRENT STATE

An 8-step manual process, repeated hundreds of times



Manual research fails quietly — it just wastes time



Repetitive

Identical steps, executed manually, for every company on the list.



Time-Consuming

Hours spent per batch that scale linearly with headcount, not tooling.



Inconsistent

Result quality depends entirely on who happens to be researching.



Low-Value Work

Skilled team members spend their time copying and pasting.



Doesn't Scale

Breaks down completely beyond a small, manageable batch size.



Error-Prone

Manual verification introduces avoidable human mistakes.

Every signal points to automation

AUTOMATION-FIT CRITERIA

 Repetitive

 Rule-Based

 Time-Consuming

 Public Data

 Performed at Scale

 Low Judgment Needed

BUSINESS VALUE DELIVERED



Reduced Research Time



Copy-Paste Work Eliminated



Reusable Founder Datasets



Consistent Data Quality

An automated discovery engine not a scraper

What It Automates

- ✓ Reads company names from Google Sheets
- ✓ Generates search queries automatically
- ✓ Searches via Google Serper API
- ✓ Extracts the most relevant founder or co-founder LinkedIn profile from search results
- ✓ Applies rule-based filtering before storing results
- ✓ Writes verified results back automatically

What It Deliberately Avoids

- ✗ Logging into LinkedIn
- ✗ Bypassing authentication
- ✗ Scraping private profiles
- ✗ Sending automated outreach
- ✗ Modifying source data unnecessarily

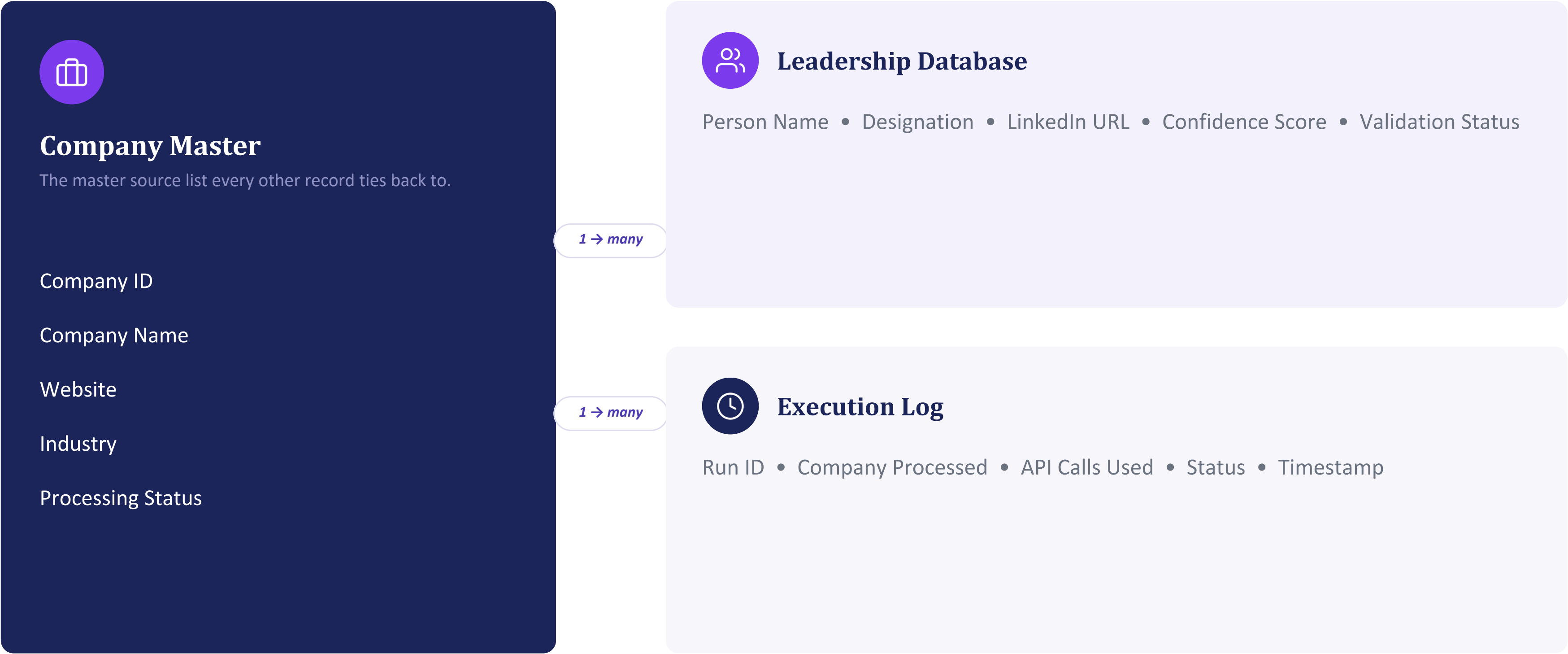
Built as a platform, not a script



Seven layers, each with one job

	Input Layer Google Sheets holds the source company list	01
	Orchestration Layer n8n reads data, calls APIs, manages retries	02
	Search Layer Serper API retrieves public results	03
	Ranking Layer Candidates are scored for relevance and confidence	04
	Validation Layer Quality control before anything is accepted	05
	Storage Layer Enriched records are written back to Google Sheets	06
	Monitoring & Logging Layer Every execution is tracked for visibility	07

Three sheets. Zero duplicate records.



One clear path from company name to verified profile

DISCOVERY PHASE

- 1 Read Company
- 2 Generate Search Query
- 3 Call Search API
- 4 Analyze Results
- 5 Identify Founder

VALIDATION & STORAGE PHASE

- 6 Extract LinkedIn URL
- 7 Validate Match
- 8 Update Google Sheet
- 9 Log Execution



Loops automatically to the next company in the sheet.

A fallback ladder, not a single guess

01 Founder + LinkedIn

02 Co-Founder + LinkedIn

03 CEO + LinkedIn

04 Leadership + LinkedIn

05 About Us / Team

Search stops the moment a valid match is found — protecting API quota and cutting response time.

Scoring Criteria

- ✓ Is the domain linkedin.com?
- ✓ Does the title say Founder or CEO?
- ✓ Does the company name match?
- ✓ Is the snippet actually relevant?

Purpose-built, not over-engineered



n8n

Workflow Automation



Google Cloud

Platform



Custom Search API

Serper APIs



Google Sheets

Database



HTTP Request

Integration



JavaScript

Processing



Cron

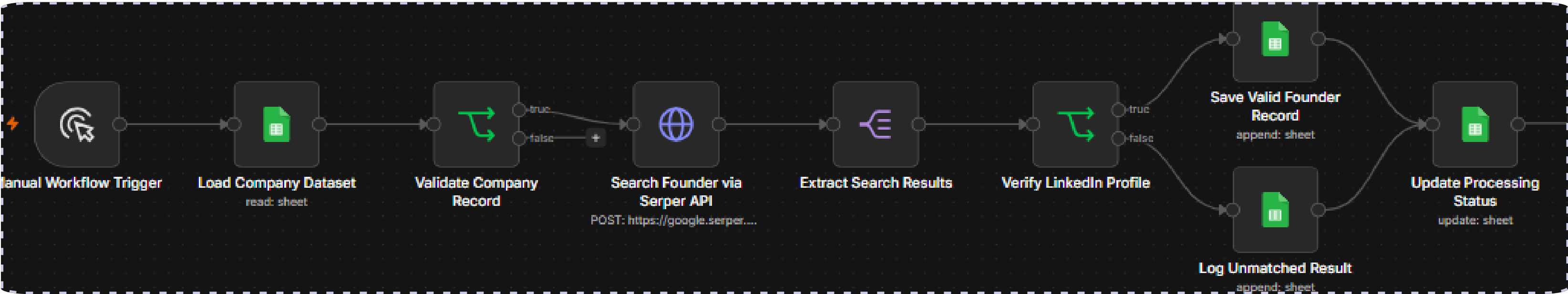
Scheduling



JSON

Data Format

Ten purpose-built nodes, one orchestration engine



1 Manual Workflow Trigger

2 Load Company Dataset

3 Validate Company Record

4 Search Founder via Serper API

5 Extract Search Results

6 Verify LinkedIn Profile

7 Ranking Node

8 Save Valid Founder Record

9 Log Unmatched Result

10 Update Processing Status

Tested module by module, not all at once

TESTING SEQUENCE



Retry Strategy

- Up to 3 retries on temporary failures
- Short delay inserted between attempts
- Logged and flagged after final retry
- Other companies keep processing



Edge Cases Tested

- Empty or misspelled company names
- Multiple founders, no clear match
- No LinkedIn result available
- Duplicate company entries

One workflow, five business functions



Recruiters

Faster identification of
founders and hiring
managers



Sales Teams

Better-researched
prospects before outreach



Business Development

Structured leadership
intelligence on demand



Investors

Faster, more consistent
startup due diligence



Job Seekers

Better understanding of
target companies

The same automated workflow serves every function above no separate tooling, no duplicated research, no re-briefing a new team on how to find the same information.

Defined, measurable success criteria



≥90%

Successful Founder Identification



<5 sec

Avg. Processing Time per Company



≥80%

Reduction in Manual Research Effort



<5%

Manual Review Rate



100%

Failure Isolation No Full Stoppage



100 → 10K+

Companies Scalable by Design

Eight versions, one architecture



Version 1 is live today. Every future version extends the same architecture new enrichment modules plug into the existing search, ranking, and validation layers without a rebuild.

What this project reinforced



Modular Workflow Design

Building components that can be replaced independently



Reliable API Integration

Working with rate limits, retries, and quotas at scale



Reusable Data Models

Structuring normalized, relationship-aware datasets



Failure as a First-Class Concern

Designing for graceful degradation, not just the happy path



Systems Thinking

Seeing the platform, not just the individual task



Business & Technical Balance

Weighing requirements against real implementation constraints



CONCLUSION

A practical solution today. A platform for tomorrow.

This project replaces repetitive manual research with a scalable, maintainable automation built on modular architecture, structured data, and graceful failure handling. It is designed to grow from founder discovery into a full company intelligence platform, without a redesign.

AI FOUNDER INTELLIGENCE & LEAD ENRICHMENT PLATFORM

Thank you.

